

Feiyang Huang

PhD Candidate, Computational Biology & Medicine

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Education

PhD	Weill Cornell Graduate School of Medical Sciences , Tri-Institutional PhD Program in Computational Biology & Medicine • Advisor: Wesley Tansey, Memorial Sloan Kettering Cancer Center	New York, NY Aug 2023 – present
BS	Johns Hopkins University , Computer Science and Biomedical Engineering • GPA 3.96 • Activities: President, Singapore Students' Association (2021–2023); Theta Tau Professional Engineering Fraternity (2020–2022)	Baltimore, MD Sept 2020 – May 2023

Experience

Memorial Sloan Kettering Cancer Center, Tansey Lab , PhD Researcher • Developing OncoState, a self-supervised transformer learning patient states from multimodal clinical trajectories of 99,000+ MSK-IMPACT patients; outperforms the IMDC risk model in advanced clear cell RCC. • Contributed to PPC , a pan-cancer atlas of ex vivo drug screens for functional precision oncology. • Contributed to a population-scale study of clinical and genetic determinants of systemic-therapy toxicity.	New York, NY June 2024 – present
Memorial Sloan Kettering Cancer Center, Tansey Lab , PhD Rotation Bayesian active learning to nominate combination therapies from ex vivo drug screens; groundwork for the PPC atlas.	New York, NY Mar 2024 – May 2024
Memorial Sloan Kettering Cancer Center, Chodera Lab , PhD Rotation Graph neural networks for small-molecule property prediction. • Integrated the ViSNet equivariant GNN into mtenn (model conversion, configuration, tests) and set up CI against stable and nightly PyTorch Geometric.	New York, NY Jan 2024 – Mar 2024
Memorial Sloan Kettering Cancer Center, CMO Innovation Lab , PhD Rotation Computational pipeline for bisulfite- and enzyme-based DNA methylation sequencing of tumor samples, with Brian Loomis.	New York, NY Nov 2023 – Jan 2024
Sloan Kettering Institute, Morris Lab , PhD Rotation BERT model for representation learning from tumor mutation profiles.	New York, NY Aug 2023 – Nov 2023
Johns Hopkins University, Fan Lab , Undergraduate Researcher • Built and benchmarked a reference-free LDA pipeline to deconvolve spatial transcriptomics data (STdeconvolve, <i>Nature Communications</i> 2022). • Developed a torque-based method and simulation framework to quantify subcellular mRNA polarization.	Baltimore, MD Jan 2021 – May 2023
Hologic , Software Engineering Intern • Built an internal work-history search tool and automated release-note generation (3 h to 20 min).	May 2022 – Aug 2022
Johns Hopkins University, Biomedical Design , Team Lead, iMEDS • Led a team of 7 building an algorithm to continuously monitor pediatric ICU sedation; interviewed 20+ clinicians.	Baltimore, MD Aug 2022 – May 2023
Johns Hopkins University, Institute for NanoBioTechnology , Research Intern • Optimized a point-of-care PCR assay (15% lower detection limit) and its embedded image-analysis and motor-control code.	Baltimore, MD May 2021 – Aug 2021
DioTeX Diagnostics , Co-Founder • Led assay development for a lateral-flow test for internal hemorrhage; raised over \$60,000 in non-dilutive funding.	Baltimore, MD Nov 2020 – Aug 2023

National University of Singapore, Yu Lab, Research Intern

- Trained a CNN for endoscopic image classification (91% accuracy); co-author in *Gastrointestinal Endoscopy*.

Singapore
Nov 2019 – Dec 2020

Singapore Armed Forces, 3rd Sergeant

- Led a detachment of 3 soldiers and co-supervised daily training for 15.

Singapore
Jan 2018 – Oct 2019

Publications**A Pan-Cancer Ex Vivo Drug Screen Atlas for Functional Precision Oncology**

Feb 2026

Karl Pichotta, Jessica B White, Jeffrey F Quinn, Anneliese Markus, Christopher Tosh, Antoine De Mathelin, Erin Coyne, **Feiyang Huang**, Wesley Tansey
pmc.ncbi.nlm.nih.gov/articles/PMC12934811 (bioRxiv (preprint))

Population-Scale Precision Safety in Oncology Reveals Clinical and Genetic Determinants of Systemic Therapy Toxicity

Sept 2026

Ziad Bakouny, X Alex Guo, **Feiyang Huang**, Saksham Mohan, Rohan Walser, et al., Wesley Tansey, Jian Carrot-Zhang, Ed Reznik
www.medrxiv.org/content/10.64898/2026.09.16.26363259 (medRxiv (preprint))

Reference-free cell type deconvolution of multi-cellular pixel-resolution spatially resolved transcriptomics data

Apr 2022

Brendan F Miller, **Feiyang Huang**, Lyla Atta, Arpan Sahoo, Jean Fan
[10.1038/s41467-022-30033-z](https://doi.org/10.1038/s41467-022-30033-z) (Nature Communications 13, 2339)

Artificial intelligence-enhanced white-light colonoscopy with attention guidance predicts colorectal cancer invasion depth

Sept 2021

Xiaobei Luo, Jiahao Wang, Zelong Han, Yang Yu, Zhenyu Chen, **Feiyang Huang**, et al., Hanry Yu
[10.1016/j.gie.2021.03.936](https://doi.org/10.1016/j.gie.2021.03.936) (Gastrointestinal Endoscopy 94(3), 627–638)

Roles and regulation of long noncoding RNAs in hepatocellular carcinoma

Oct 2019

Lee Jin Lim, Samuel YS Wong, **Feiyang Huang**, et al., Caroline G Lee
[10.1158/0008-5472.CAN-19-0255](https://doi.org/10.1158/0008-5472.CAN-19-0255) (Cancer Research 79(20), 5131–5139)

Conference Abstracts

- Bakouny Z, Guo XA, Walser R, **Huang F**, et al. Pan-cancer discovery of clinical and genomic determinants of immune-related adverse events. *Journal of Clinical Oncology* 44(16_suppl):11132 (ASCO 2026).
- Wu S, Pejic B, Roy S, **Huang F**, et al. Development of a computational model of SBS scores in critically ill children. *Pediatric Critical Care Medicine* 25(11S):e87 (2024).

Patents

- Parent C, Simon E, **Huang F**, Ruci A, Zhang E. Methods and devices for quantitatively estimating thrombomodulin. US Patent App. 19/240,058 (2025).
- Eng R, Parent C, Hayat A, Ruci A, Krishnan A, **Huang F**, Simon E, Zhang E. Methods and devices for quantitatively estimating syndecan-1. US Patent App. 17/715,294 (2022).

Presentations

- 45th Vincent du Vigneaud Research Symposium, Weill Cornell Medicine, New York, NY (2026). **Huang F**, Tansey W. OncoState: learning patient states from clinical trajectories in cancer. Poster #66.
- Military Health System Research Symposium, Kissimmee, FL (2022). Towards point-of-care internal hemorrhage detection through a syndecan-1 lateral flow assay. Oral presentation.
- Johns Hopkins BME Design Day, Baltimore, MD (2022). DioTeX: Hemorrhage Diagnostics. Poster.

Honors and Awards**NIH DEBUT Challenge, Healthcare Technologies for Low-Resource Settings Prize**

2023

With the DioTeX team; \$15,000.

VentureWell e-Fest Prize

2023

With the DioTeX team; \$10,000.

Teaching

- Course Assistant, EN.601.230 Mathematical Foundations for Computer Science, Johns Hopkins University (Fall 2022).